

Q. 1 Prove that, for any natural number n , there exists a n digit number which contains only the digits 2 and 3, and is divisible by 2^n .

Q. 2 For all natural numbers n , define the polynomial

$$P_n(x) := x^{n+2} - 2x + 1.$$

Then

(a) Show that the equation $P_n(x) = 0$ has one and only one root c_n in $(0, 1)$.

(b) Calculate $\lim_{n \rightarrow \infty} c_n$.

Q. 3 Show that if $a \neq b$, then

$$\frac{e^b - e^a}{b - a} < \frac{e^a + e^b}{2}.$$

Q. 4 Prove that there cannot exist any function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for every $n \geq 2$,

$$f(f(n-1)) = f(n+1) - f(n)$$

holds.

Q. 5 Suppose that a polynomial $P(x) = x^{10} - 10x^9 + 39x^8 + a_7x^7 + \cdots + a_1x + a_0$ has 10 real roots. Prove that all the roots of $P(x)$ lie between -2.5 and 4.5 .

Q. 6 Let a sequence of real numbers $\{a_n\}_{n \geq 1}$ be defined as

$$a_n := \sum_{k=1}^n \frac{1}{k^3}.$$

Prove that

$$\sum_{k=1}^n \frac{1}{k^3 a_k^2} < \frac{6}{5}.$$

Q. 7 Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuously differentiable function such that

$$\int_0^1 f(x) dx = 1.$$

Prove that

$$\int_0^1 (1+x^2)f^2(x) dx \geq \frac{4}{\pi}.$$

Q. 8 Suppose that points P and Q lie on side BC of acute-angled $\triangle ABC$ such that $\angle PAB = \angle ACB$ and $\angle CAQ = \angle ABC$. Let M and N be the points on lines AP and AQ , respectively, such that P is the midpoint of AM , and Q is the midpoint of AN . Prove that the intersection of BM and CN lies on the circumcircle of $\triangle ABC$.

-End of the Question Paper-